

Lesson 3: Layout — Row, Column & Container

FlutterLearn | Module 1: Flutter Foundations

How Flutter lays things out

Flutter layout is beautifully simple: widgets are arranged by parent widgets whose only job is positioning. Want things side by side? Put them in a Row. Stacked vertically? A Column. Need padding or a background color? Wrap it in a Container.

You rarely position things with x/y coordinates. You compose layout widgets, and Flutter measures everything for you.

Row and Column

Row arranges its children horizontally. Column arranges them vertically. Both take a children list:

```
Column(  
  children: const [  
    Text('First'),  
    Text('Second'),  
    Text('Third'),  
  ],  
)
```

Two properties control alignment, and they're worth memorizing:

- `mainAxisAlignment` — alignment along the main direction (horizontal for Row, vertical for Column).
- `crossAxisAlignment` — alignment across it.

So in a Column, `mainAxisAlignment: MainAxisAlignment.center` centers children vertically, while `crossAxisAlignment: CrossAxisAlignment.center` centers them horizontally. Common values: `start`, `center`, `end`, `spaceBetween`, `spaceEvenly`.

```
Row(  
  mainAxisAlignment: MainAxisAlignment.spaceEvenly,  
  children: const [  
    Icon(Icons.star),  
    Icon(Icons.favorite),  
    Icon(Icons.thumb_up),  
  ],  
)
```

Container: the styling workhorse

Container is the widget you reach for when you need padding, margins, background color, borders, or a fixed size:

```
Container(  
  padding: const EdgeInsets.all(16),  
  margin: const EdgeInsets.symmetric(vertical: 8),  
  color: Colors.blue,  
  child: const Text(  
    'Inside a container',  
    style: TextStyle(color: Colors.white),  
  ),  
)
```

`EdgeInsets.all(16)` means 16 pixels on every side. `EdgeInsets.symmetric` lets you set vertical and horizontal separately. You'll type `EdgeInsets` a hundred times — it becomes muscle memory fast.

Putting it together: a simple card

```

Container(
  padding: const EdgeInsets.all(16),
  margin: const EdgeInsets.all(16),
  color: Colors.white,
  child: Row(
    children: const [
      Icon(Icons.person, size: 48),
      SizedBox(width: 16),
      Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          Text('Aarav Sharma', style: TextStyle(fontSize: 20)),
          Text('Flutter Learner'),
        ],
      ),
    ],
  ),
)

```

A Container holds a Row; the Row holds an icon, a SizedBox spacer, and a Column of two texts. Layout widgets nesting inside layout widgets — that's 90% of real Flutter UI work.

The classic beginner error: overflow

If a Row's children are wider than the screen, Flutter shows the infamous yellow-and-black striped overflow warning. The fix is usually wrapping children in Expanded, which tells a child to take up the remaining space:

```

Row(
  children: const [
    Expanded(child: Text('This text takes all remaining space')),
    Icon(Icons.arrow_forward),
  ],
)

```

You'll meet this bug in the exercise below — on purpose.

Key terms

- Main axis — the direction a Row/Column lays children out in.
- Cross axis — the direction perpendicular to the main axis.
- Expanded — forces a child of a Row/Column to fill remaining space.

- EdgeInsets — describes padding and margins.